

BONE PATHOLOGY I

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HEREDITARY BONE DISORDERS

A. Achondroplasia

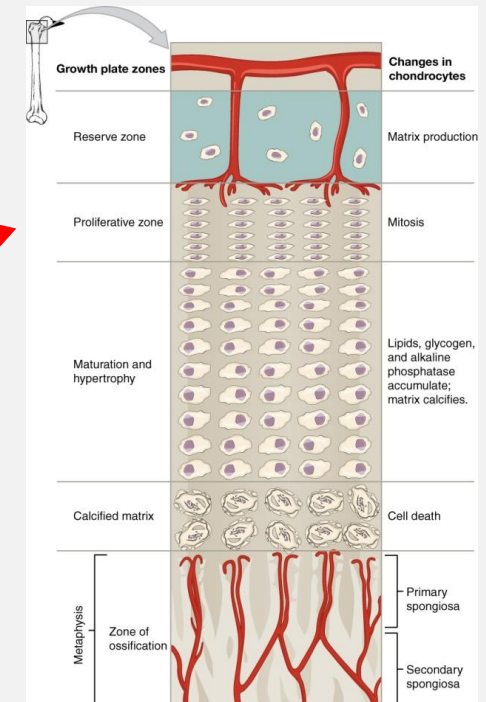
- Most common form of **inherited dwarfism**
- **(AD)**-Autosomal dominant (80% sporadic new mutations)
- Disorder of cartilage proliferation
- Mutation is : Overactive FGFR3 **inhibits** cartilage proliferation at epiphyseal growth plate, resulting in decreased endochondral bone formation



Pathogenesis

- During embryonic establishment of the growth plate before formation of the secondary ossification center, **FGFR3 signaling promotes chondrocyte proliferation**
- However, during postnatal skeletal growth, *FGFR3 signaling* **inhibits** chondrocyte proliferation and differentiation
- *In Achondroplasia FGFR3 is **constitutively active***

Activated
FGFR3
**Inhibits
proliferation**

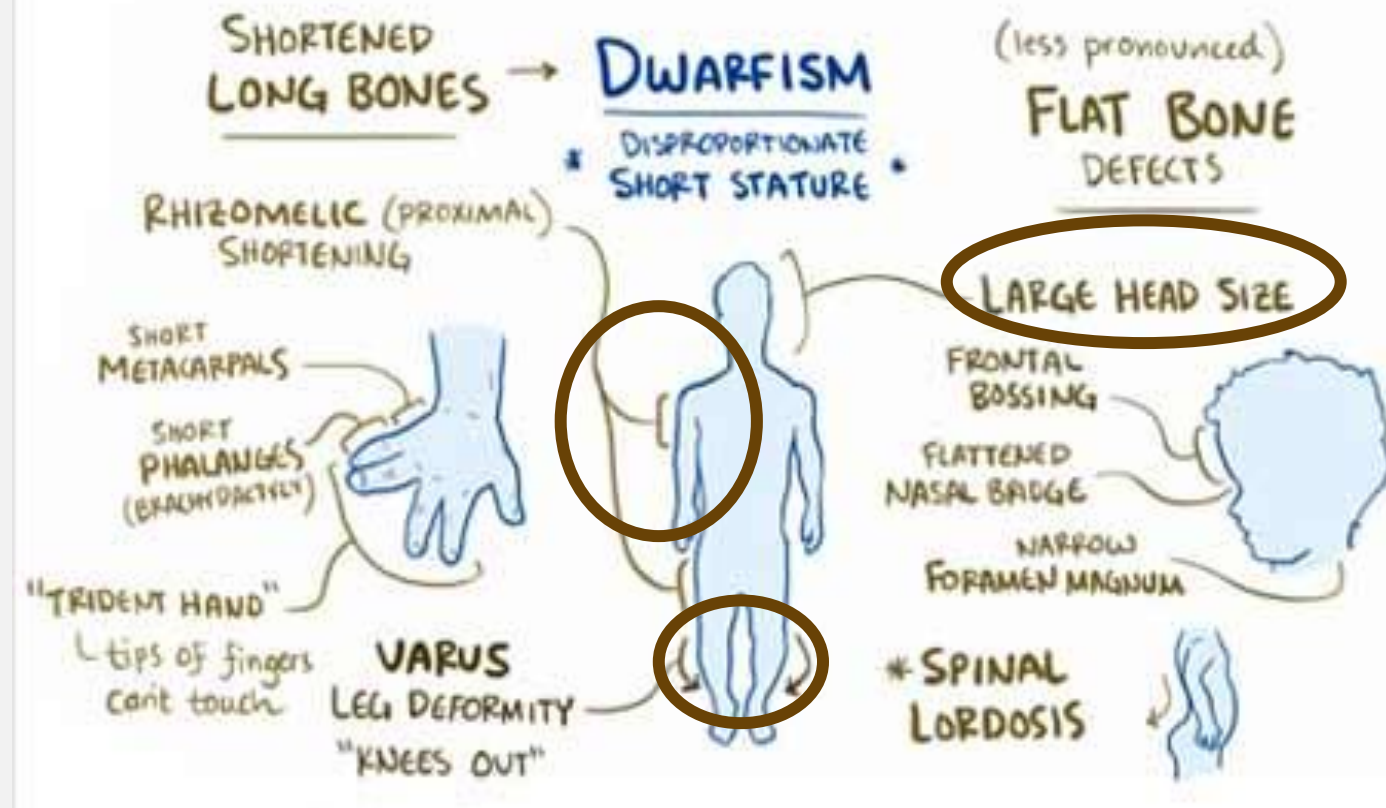


Diagnosis

The diagnosis of Ach is usually made at birth

- **short long bones**, disproportional shortening of the proximal skeletal segments (**rhizomelia**),
- **tibial bowing**, exaggerated lumbar lordosis, ,
- **macrocephaly**,
- **hearing loss**

Symptoms of Achondroplasia



B. Osteogenesis imperfecta (brittle bone disease)

- Autosomal Dominant
- Abnormal synthesis of **type I collagen and bone mineralization**
- Increased susceptibility to bone fractures
- Decreased bone density.

Skeletal:
short stature,
scoliosis,
skull deformities



Extraskeletal:
hearing loss
sclera may be blue
joint hyperextensibility

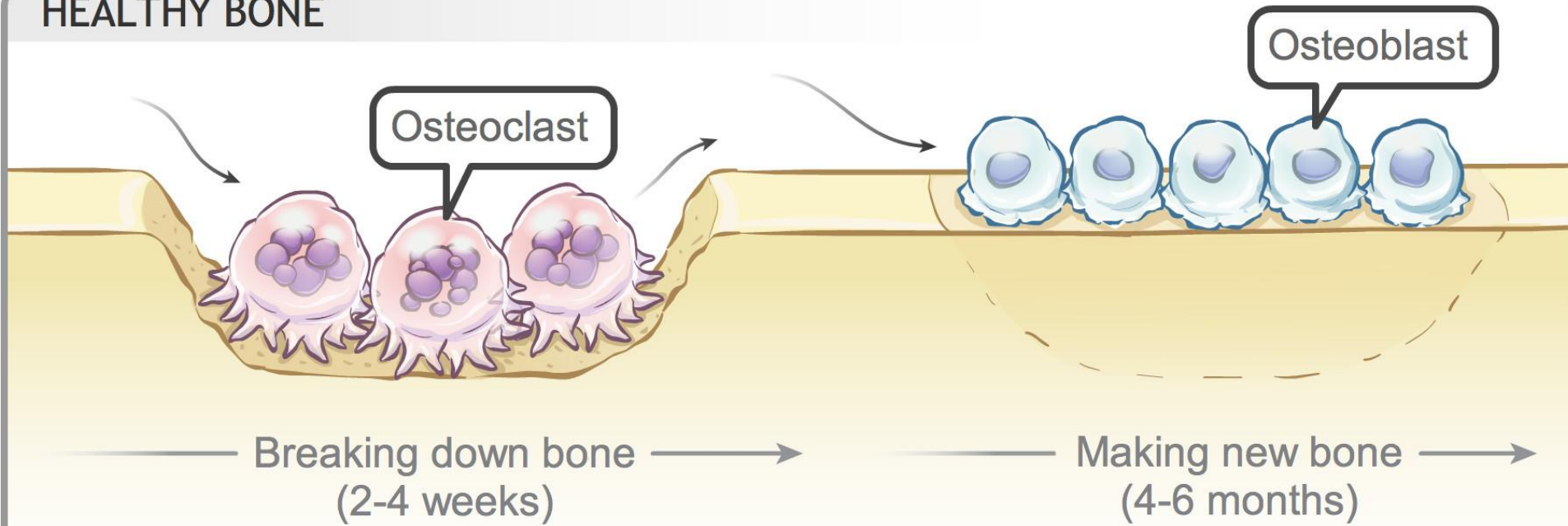
C. Osteopetrosis

Osteopetrosis is a heterogeneous group of **heritable conditions** in which there is a **defect in bone resorption by osteoclasts**.

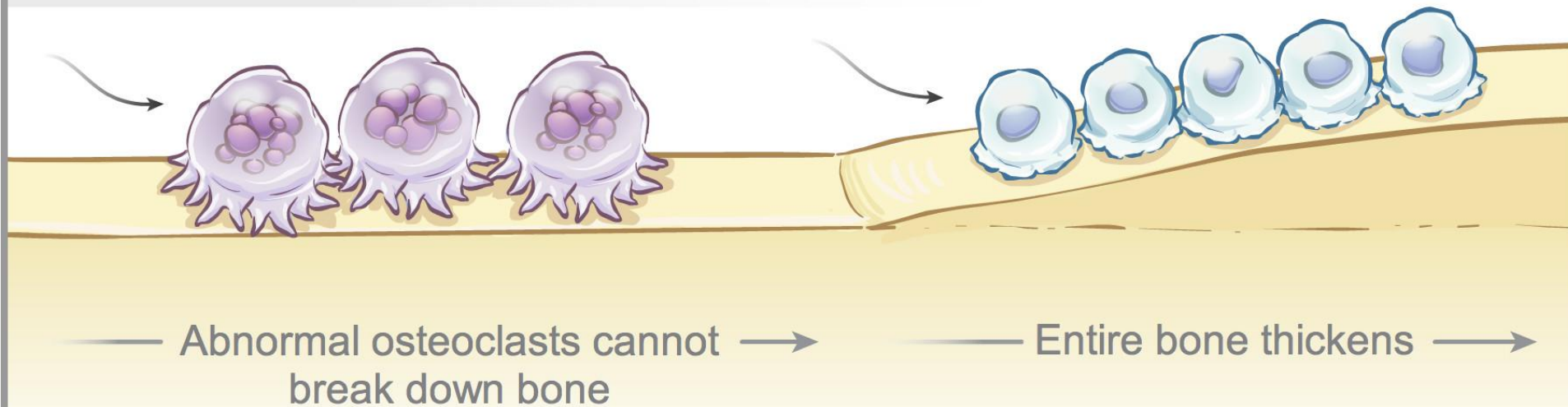
Bone is a **dynamic tissue** in which osteoblasts synthesize bone matrix while *osteoclasts resorb bone*.

Bone **density** is dependent on the relative function of these two types of cells. Osteoclasts and Osteoblasts

HEALTHY BONE



OSTEOPETROTIC BONE



Pathology:

- Impaired **osteoclast's** function,
- Decreased **resorption** leads to **thick sclerotic bones**
- Interference of **acidification** by osteoclasts , required for bone resorption.
- Defective gene which encodes for **carbonic anhydrase II** and **chloride channel gene** (*acidic environment is necessary to remove bone*)



Pathogenesis and basics in depth:

Osteoclast precursors are of **monocyte lineage** which differentiate into osteoclasts. Osteoclasts become adherent to bone.

A tight seal of attachment between podocyte of osteoclast and bone surface.

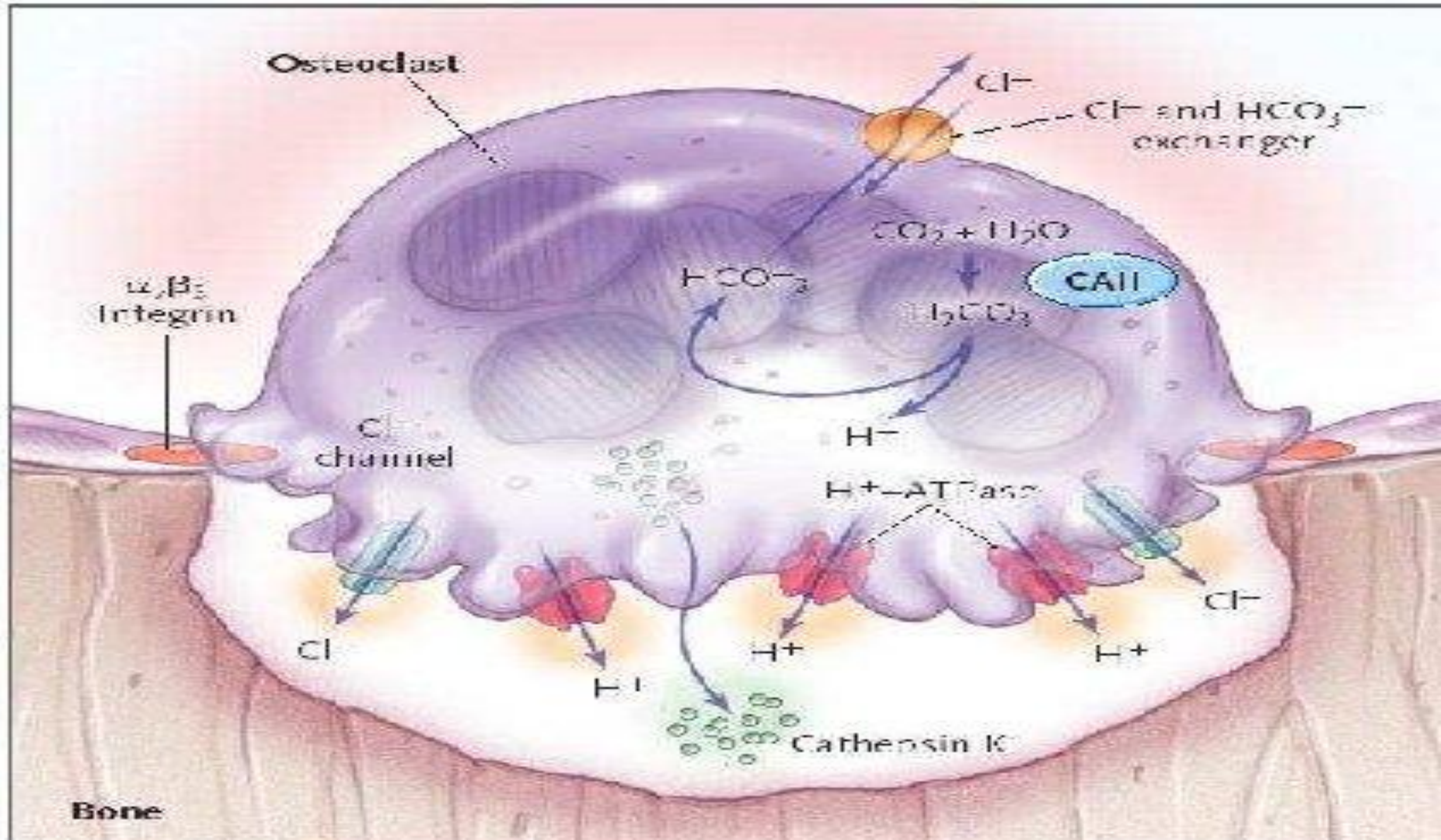
Polarization of the osteoclast forms two functionally distinct domains, the resorptive surface and the basolateral membrane. The “**ruffled border**,” is the resorptive surface of the cell.

Resorption occurs through **acidification of the bony surface**, which initiates dissolution of the mineral matrix and secretion of enzymes that digest the organic component of bone.

In the **cytoplasm of the osteoclast**, **carbonic anhydrase II** forms carbonic acid (H_2CO_3) from carbon dioxide (CO_2) and water; the H_2CO_3 dissociates to bicarbonate (HCO_2^-) and a proton (H^+).

The protons are transported through the ruffled border into the resorption lacuna by a vacuolar proton pump (H^+-ATPase), generating a pH of 4 to 5 in the extracellular space adjacent to bone.

Acidification of this extracellular environment initiates the degradation of the mineral component of bone, which is composed primarily of hydroxyapatite



HEREDITARY DISORDERS OF BONE

ACHONDROPLASIA	OSTEOGENESIS IMPERFECTA	OSTEOPETROSIS
decreased endochondral bone formation	Abnormal synthesis of type I collagen and bone mineralization	Impaired osteoclasts